Type MS · Convert units and write numbers in standard form or significant figures

Answer 1 of 4 · Question 3 part a.ii · 2 marks

2022 JUN · Paper 01 (regional R) · extract 1 of 2 · find part a.ii in this mark scheme extract

Number				
3 (a)	(i)	3 320 000;		1
	(ii)	6.00239; x 10^5;	Allow 6.0024	2
	(iii)	pancreas;		1
(b)	(i)	120 ÷ 1.95^2/3.8029; 31.6;	Allow figure not rounded - 31.5581 Allow 31.56/32	2
	(ii)	obese;	ecf from previous question	1
(c)	(i)	• as BMI increases the risk of diabetes increases; • between a BMI of 24-35 the risk of diabetes increases more rapidly/BMI less than 25 the risk increases less rapidly;	Allow any range values between 24-35	2
	(ii)	Any two from the following: • balanced diet; • eat less fat; • eat less sugar	Ignore healthy diet Allow reduce alcohol/carbohydrate consumption for mp3	2

Total marks 11

Type MS · Convert units and write numbers in standard form or significant figures

Answer 1 of 4 · Question 3 part a.ii · 2 marks

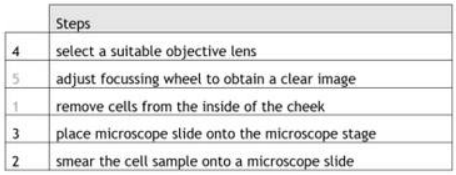
2022 JUN · Paper 01 (regional R) · extract 2 of 2 · find part a.ii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Convert units and write numbers in standard form or significant figures

Answer 2 of 4 · Question 3 part b.ii · 2 marks

2020 NOV · Paper 01 (regional R) · find part b.ii in this mark scheme extract

Number				
3	(a) (i)		One mark for 4 before 5. One mark for 3 before 2	2
	(ii)	more detail/organelles can be seen/organelles/named organelle seen more clearly (1)		1
	(iii)	reference no stain applied to specimen(1)		1
	(iv)	• reference to sterilising (named) equipment (1) • wash hands to avoid cross contamination(1) • incubate/store in sealed petri dish / container	allow valid sterilisation technique	2
	(b) (i)	10 × 40 = 400	Full marks for correct answer without working	1
	(ii)	reference to 1000 (1) 0.018 × 1000 = 18 (µm)	Full marks for correct final answer	2

Total Question 3 = 9 marks

Type MS · Convert units and write numbers in standard form or significant figures

Answer 3 of 4 · Question 3 part a.i · 1 mark

2022 JUN · Paper 01 (regional R) · extract 1 of 2 · find part a.i in this mark scheme extract

number				
3 (a)	(i)	3 320 000;		1
	(ii)	6.00239; x 10^5;	Allow 6.0024	2
	(iii)	pancreas;		1
(b)	(i)	120 ÷ 1.95^2/3.8029; 31.6;	Allow figure not rounded - 31.5581 Allow 31.56/32	2
	(ii)	obese;	ecf from previous question	1
(c)	(i)	• as BMI increases the risk of diabetes increases; • between a BMI of 24-35 the risk of diabetes increases more rapidly/BMI less than 25 the risk increases less rapidly;	Allow any range values between 24-35	2
	(ii)	Any two from the following: • balanced diet; • eat less fat; • eat less sugar	Ignore healthy diet Allow reduce alcohol/carbohydrate consumption for mp3	2

Total marks 11

Type MS · Convert units and write numbers in standard form or significant figures

Answer 3 of 4 · Question 3 part a.i · 1 mark

2022 JUN · Paper 01 (regional R) · extract 2 of 2 · find part a.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Convert units and write numbers in standard form or significant figures

Answer 4 of 4 · Question 3 part a.ii · 1 mark

2023 JAN · Paper 02 · extract 1 of 2 · find part a.ii in this mark scheme extract

Number				

3	(a) (i)	surface area = 24 (mm^2);
		volume = 8 (mm^3);
		3:1;
			Full marks for correct final answer	3

	(ii)	C 92;		1
	(iii)	movement from a high concentration to a low concentration;		1
	(iv)	• reduces the time taken to become colourless/rate of diffusion/reaction is faster; • the diffusion distance is shorter/greater area over which diffusion/collisions can occur;		2
	(b) (i)	• use acid heated to different temperatures/idea of different temperatures; • compare/measure the time taken to become colourless/compare rate of diffusion;		2
	(ii)	Any two from: • the side length of the cube/size of the cube/surface area/volume of cube; • acid concentration; • concentration of alkali/indicator in the cube;	accept volume of acid Allow volume of indicator	2

Total 11 marks

Type MS · Convert units and write numbers in standard form or significant figures

Answer 4 of 4 · Question 3 part a.ii · 1 mark

2023 JAN · Paper 02 · extract 2 of 2 · find part a.ii in this mark scheme extract

Question number	Answer	Notes	Marks